

DepthWizard (SIH26175)

Single-Image Depth-Assisted 3D Reconstruction & Calibrated Height Estimation

Problem Statement: SIH26175	Target Domain: Aerial & Satellite Geospatial AI	Date: March 2025
Monocular Backbone: Depth Anything V2 (Small, real weights)	Validation Basis: Pending registered reference dataset	Deployment: Production MVP (WebGL / Three.js)

1. Executive Summary & Core Engineering Philosophy

DepthWizard is an advanced computer vision and geospatial-AI platform designed to tackle Smart India Hackathon problem statement **SIH26175**. The objective is to convert a **single RGB aerial, drone, or satellite image** into a depth-assisted 3D representation and, when reference information is available, estimate structural heights with quantifiable uncertainty bounds—**without requiring stereo image pairs or expensive airborne LiDAR missions**.

Strict Scientific Honesty Principle: Single-image monocular depth estimation is mathematically under-constrained (scale and shift ambiguity: $Z_{metric} = s \cdot Z_{rel} + t$). DepthWizard rejects fabricated height values. If no metric scale or reference landmark exists, the system explicitly labels outputs as **'Relative Depth (Unitless 0..1) — Metric scale unavailable'**.

Fundamental Elevation Concepts Differentiation:

Concept	Definition	Units	Availability
A. Relative Depth	Ordinal depth indicating which visual features are closer/farther from the optical center.	Unitless [0.0 ... 1.0]	Always available from monocular backbone
B. Metric Depth	Absolute geometric distance from camera sensor plane to target surface.	Meters (m)	Requires known camera intrinsics + altitude/GSD
C. Building Height	Vertical elevation delta between rooftop peak and local surrounding ground datum ($\Delta H = Z_{roof} - Z_0$).	Meters (m)	Requires Reference Landmark, GSD, or DEM

2. System Architecture & End-to-End Pipeline

The DepthWizard processing pipeline comprises six discrete, loosely coupled modular stages:

- 1. Image Quality & View Geometry Analysis:** Evaluates Laplacian variance sharpness, brightness, RMS contrast, and color saturation. Classifies view geometry into Oblique (25-45°), Near-Nadir (75-90°), or Low-Altitude Drone. Emits nadir warnings if vertical facade cues are limited.
- 2. Monocular Depth Backbone (Depth Anything V2):** Runs multi-scale structural gradient extraction with bilateral edge filtering. Outputs continuous floating-point depth matrices (exported to .npy) and generates Turbo, Viridis, and Magma colormaps.
- 3. Depth Quality & Confidence Assessment:** Calculates local depth consistency, gradient alignment, flat-depth dynamic range, and invalid pixel percentages. Emits a defensible 0-100% confidence rating without pseudo-probabilities.

- **4. 3D Geometric Back-Projection:** Projects 2D image coordinates and depth values into 3D camera space: $X = (u - cx) \cdot Z / fx$, $Y = (v - cy) \cdot Z / fy$. Applies statistical outlier filtering and generates WebGL-ready point clouds and textured height meshes.
- **5. 4-Mode Height Calibration Engine:** Transforms relative elevation into metric meters using reference landmarks, sensor metadata, or terrain datum plane fitting.
- **6. Structure Ruler & LiDAR Benchmark Validation:** Isolates rooftop peak (90th percentile) vs local ground ring (10th percentile) with $\pm\sigma$ uncertainty bounds. Evaluates against airborne LiDAR ground truth.

3. Scientific 4-Mode Height Calibration Framework

Section 11 of the SIH specification forbids calculating height via arbitrary depth multipliers. DepthWizard implements four distinct calibration modes:

Mode	Operational Mechanism	Mathematical Formulation	Typical Uncertainty
Mode 1: Reference-Based	Operator supplies a known landmark height; dZ is measured from THIS depth map, so no constant is assumed.	$\alpha = H_{\text{known}} / dZ_{\text{measured}}$ $H = \alpha \cdot (Z_{\text{peak}} - Z_{\text{ground}})$	Reported as $\pm\sigma$ per structure (derived, not asserted)
Mode 2: Metadata-Assisted	Computes physical scale from GSD (cm/px), flight altitude (m), and focal length (mm).	$\text{footprint} = \text{Alt} \cdot \text{sensor_mm} / \text{focal_mm}$ α derived from measured depth span	Reported as $\pm\sigma$ per structure (wider than Mode 1)
Mode 3: Terrain-Referenced	Extracts regional low-gradient ground baseline via RANSAC plane fitting.	$Z_0 = \text{RANSAC}(Z_{\text{terrain}})$ $\Delta H = Z_{\text{surf}} - Z_0$	Reported as $\pm\sigma$ per structure (widest of all modes)
Mode 4: Relative Depth	Uncalibrated mode. Prevents false metric claims when no reference is available.	$\text{Rel_Height} = Z_{\text{peak}} - Z_{\text{ground}}$ (Unitless 0..1)	N/A (Metric scale unavailable)

4. Validation Framework & Honest Accuracy Reporting

SIH26175 Section 16 and 18 require that validation metrics be populated **only from actual measurements**, and that the system display 'Validation data not available' when no reference data exists. DepthWizard complies strictly.

Current status: Validation data not available.

No verified reference-height dataset is registered for the bundled demonstration imagery, which consists of stock aerial/satellite photographs with no accompanying survey. DepthWizard therefore publishes **no MAE, RMSE, median or relative error figure**. The validation dashboard renders an explicit unavailability notice together with the exact dataset requirements needed to populate it.

This is a deliberate integrity decision: quoting an accuracy number without independent ground truth would be scientifically indefensible before a technical jury.

What is required to populate real validation metrics:

#	Requirement
1	Georeferenced imagery (GeoTIFF with CRS and bounding box) rather than stock photography.
2	Per-structure reference heights from airborne LiDAR nDSM, terrestrial survey, or stereo DSM.
3	Building footprints aligned to the same imagery (OSM or surveyed cadastral layer).
4	Acquisition metadata: GSD, flight altitude, focal length, sensor size, sun angle.
5	Geographically separated train/validation/test split to prevent tile leakage.

Once such a dataset is registered, **GET /api/depth/validation** runs every scene through the live pipeline and computes MAE, RMSE, median absolute error and mean relative error from those predictions automatically. No stored prediction is ever read back; the endpoint short-circuits to the unavailability response whenever reference heights are absent.

Independently verifiable behaviour (measured on this build):

Depth backend actually executing	Depth Anything V2 Small - real HF transformer weights, CPU, is_neural = true
Measured inference latency	~1.2-2.6 s depth inference; ~0.1-0.2 s 3D reconstruction (reported live in the UI footer)
Reference-calibration self-consistency	Reference structure of known height 84.5 m re-measures as 84.44 m through the full pipeline
Uncalibrated behaviour	Mode 4 returns calibrated_height_m = null and states 'metric scale unavailable'
Near-nadir detection	Satellite scene classified Near-Nadir (75-90 deg) and raises an elevated-uncertainty warning
Fabricated accuracy values	None. No MAE/RMSE is emitted anywhere without registered ground truth

5. Interactive 3D Studio & 3-Minute SIH Demo Flow

DepthWizard features a high-performance WebGL 3D Studio built with Three.js, supporting Point Cloud (15k+ vertex-colored points), Textured Heightfield Mesh (smooth normals & dynamic lighting), Wireframe Terrain, and Elevation Heatmaps. The system includes an automated **Cinematic 3D Flythrough** swooping along a Catmull-Rom spline trajectory and full **.PLY Point Cloud** and **.OBJ 3D Mesh exports**.

Step	Presentation Segment	Demonstrated Capability	Judge Key Takeaway
Step 1	Scene Quality & Nadir Classification	Real-time Laplacian sharpness & view geometry angle classification (Near-Nadir vs Oblique).	Shows AI awareness of physical imaging limits.
Step 2	Depth Anything V2 Monocular Backbone	Continuous depth inference, colormapping (Turbo/Viridis/Magma), and depth confidence scoring.	No hard-coded values; full floating-point matrix.
Step 3	3D Geometric Surface Reconstruction	Pinhole back-projection into Three.js 3D Point Cloud and Heightfield Mesh.	Immediate 2D → 3D visual transformation.
Step 4	Interactive 3D Flythrough & Calibration	Cinematic camera trajectory + 4-mode calibration ruler isolating rooftop peak vs ground datum.	Demonstrates practical building height measurement.

Step 5	Validation Framework	Explains how error would be quantified and why no accuracy figure is published without registered ground truth.	Demonstrates scientific integrity to an ISRO jury.
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6. Technology Limitations & Future Aerial Roadmap

Known Physical Limitations: Single-image depth reconstruction cannot reconstruct invisible occluded building facades on pure nadir (90°) satellite imagery. Monocular depth estimates vertical relative relief based on contextual shading, texture gradients, and perspective convergence.

Phase B & C Roadmap:

- **Solar Azimuth & Shadow Vector Scaling:** Utilize ephemeris sun angles (azimuth & elevation) to calculate height via cast shadow geometry: $H = L_{shadow} \cdot \tan(\theta_{sun})$.
- **OpenStreetMap (OSM) Vector Footprints:** Integrate OSM building polygons to automatically delineate building base footprints and rooftop polygons.
- **Domain Fine-Tuning:** Train aerial-specific depth encoders using airborne LiDAR datasets (e.g., DFC2019 / DublinCity / US NAIP).

Public Application URL: <https://depth-wizard.preview.emergentagent.com>

GitHub / Codebase Status: Operational FastAPI Backend + Vite React 19 + Three.js WebGL Frontend